Assignment 17

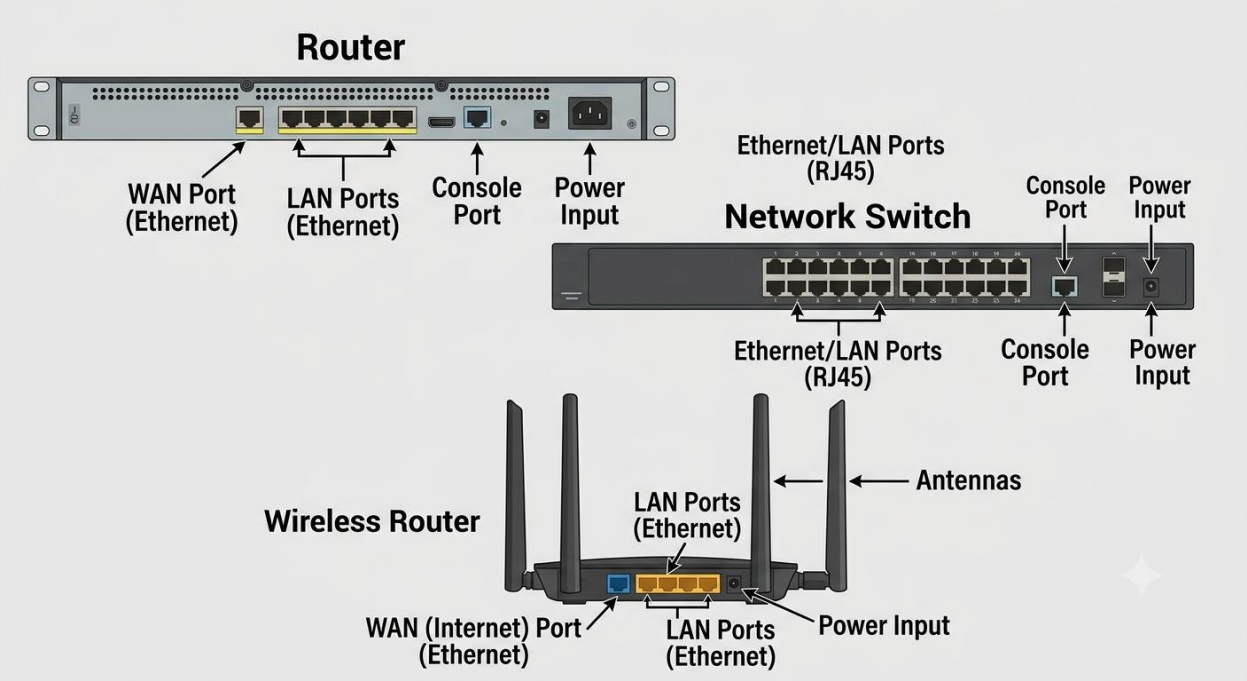
Introduction to Network Devices Theory

1. **Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.**

Device	Main Function	Typical Ports / Connections
Switch	Connects wired devices like PCs and printers in a local network.	Ethernet ports (LAN)
Router	Connects the home network to the internet and forwards data between networks.	WAN port and LAN ports
Hub	Connects multiple wired devices and sends data to all connected devices.	Ethernet ports
Firewall	Protects the network by blocking unwanted or unsafe traffic.	WAN/LAN interfaces; uses rules for ports like 80 (HTTP) and 443 (HTTPS)
Access Point (AP)	Connects wireless devices like phones and laptops to the network.	Ethernet uplink port and Wi-Fi radio

- **Note:** Switches, routers, hubs, and access points have physical ports. Firewall port numbers such as 80 and 443 are logical network ports used by applications, not physical connectors.

2. **Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.**



3. Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

❖ **I use Instagram daily to view photos and videos.**

1. The **access point** connects my phone to the home Wi-Fi network.
2. The **router** sends my data to the internet and brings Instagram's content back to my phone.
3. The **firewall** helps protect **Instagram's servers** by blocking unwanted network traffic.
4. These devices work together to help me use Instagram smoothly and safely.

4. **Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.**

- **One possible reason is that the college access points cannot handle too many students connected at the same time. This can cause slow Wi-Fi and frequent disconnections. Upgrading to higher-capacity access points with better coverage would help support more students and provide a faster, more stable connection.**